Classification - Confusion Metrics

Model-wise Questions and Answers with Final Conclusion

1. Random Forest

S.No	Question	Answer
1	What is the overall accuracy of the model?	**92%**
2	How many test samples were correctly classified?	**110**
3	How many test samples were incorrectly classified?	**10**
4	How many total samples were used for testing?	**120**
5	Which class has the higher precision score?	**Class 0 (0.95)**
6	Which class has the higher recall score?	**Class 0 (0.92)**
7	Which class has the higher F1-score?	**Class 0 (0.94)**
8	How many Class 0 samples were correctly classified?	**73**
9	How many Class 1 samples were correctly classified?	**37**
10	What is the weighted average F1-score?	**0.92**

2. Decision Tree

S.No	Question	Answer
1	What is the accuracy of the Decision Tree model?	**90%**
2	How many correct predictions are made?	**108**
3	How many incorrect predictions are made?	**12**
4	What is the total number of test samples?	**120**
5	What is the precision of Class 0?	**0.91**
6	What is the recall of Class 1?	**0.83**
7	What is the F1-score of Class 1?	**0.85**
8	How many Class 0 samples are correctly predicted?	**74**
9	How many Class 1 samples are correctly predicted?	**34**
10	What is the weighted average precision?	**0.90**

3. Support Vector Machine (SVM)

S.No	Question	Answer
1	Which model is used for classification?	**SVM**
2	What type of kernel is used in the model?	**Linear Kernel**
3	What is the highest recall value?	**0.97**
4	Which class has the lowest recall?	**Class 1 (0.41)**
5	What is the precision value of Class 1?	**0.89**
6	What is the macro average F1-score?	**0.71**
7	How many Class 0 samples are incorrectly classified?	**2**
8	How many Class 1 samples are incorrectly classified?	**24**
9	What is the F1-score of Class 1?	**0.57**
10	What is the overall model accuracy?	**78%**

4. Logistic Regression (Sigmoid)

S.No	Question	Answer
1	How reliable is the model in making predictions?	**89% Accuracy**
2	Which category is identified more successfully?	**Class 0**
3	How many predictions did the model get right?	**107**
4	Which category needs better prediction improvement?	**Class 1**
5	How many actual Class 0 samples were missed?	**5**
6	How many actual Class 1 samples were identified correctly?	**33**
7	Which category gives a better balance between precision and recall?	**Class 0 (F1-score: 0.92)**
8	How many predictions were made incorrectly in total?	**13**
9	How consistent is the model across both classes?	**Macro F1-score: 0.88**
10	What is the model's overall balanced performance?	**Weighted F1-score: 0.89**

5. K-Nearest Neighbors (KNN)

S.No	Question	Answer
1	If 120 samples are tested, how many are classified correctly?	**101**
2	Which class is more difficult for the model to identify?	**Class 1**
3	What is the difference between the recall scores of both classes?	**0.35**
4	How many Class 0 samples are predicted as Class 1?	**3**
5	Which class gives more reliable positive predictions?	**Class 1 (Precision: 0.89)**
6	What percentage of all test samples are classified incorrectly?	**16%**
7	Which class performs better in overall classification?	**Class 0 (F1-score: 0.89)**
8	How many actual Class 1 samples are identified correctly?	**25**
9	What is the average F1-score of both classes?	**0.81**
10	How much accuracy does the model achieve?	**84%**

6. Naive Bayes

S.No	Question	Answer
1	Which classification category shows the lowest detection rate?	**Class 1**
2	What proportion of Class 0 predictions are incorrect?	**18%**
3	Which class has a higher balance between precision and recall?	**Class 0**
4	How many samples are wrongly assigned to Class 0?	**18**
5	What does the model's 0.84 accuracy score indicate?	**84% correct predictions**
6	Which class has a precision advantage over the other?	**Class 1**
7	How many samples are incorrectly classified in total?	**21**
8	What is the recall performance gap between Class 0 and Class 1?	**0.33**
9	Which evaluation score indicates the harmonic balance of two metrics?	**F1-score**
10	What is the total number of samples used for evaluation?	**134**

Model Comparison Summary

Rank	Model	Accuracy	Correct / Total	Weak Point
1	Random Forest	92%	110 / 120	Class 1 slightly lower than Class 0
2	Decision Tree	90%	108 / 120	Class 1 recall 0.83
3	Logistic Regression	89%	107 / 120	Class 1 needs improvement
4	KNN	84%	101 / 120	Class 1 recall gap of 0.35
4	Naive Bayes	84%	113 / 134*	Class 1 lowest detection rate
6	SVM (Linear)	78%	94 / 120	Class 1 recall only 0.41

* Naive Bayes was evaluated on 134 samples (21 incorrect), while the other models used 120 test samples, so its comparison is approximate. SVM correct count = 120 minus 26 errors.

Final Conclusion

Best Algorithm for this Dataset: Random Forest
It gives the highest accuracy (92%, 110 out of 120 correct) and the highest weighted F1-score (0.92) among all the models.